CO_2 Reforming of *n*-Heptane on a Ni/Al_2O_3 Catalyst

K. Johanna Puolakka, A. Outi I. Krause

Helsinki University of Technology, Department of Chemical Technology,
P.O.Box 6100, FIN-02015 Helsinki University of Technology, Finland
E-mail: puolakka@polte.hut.fi

Introduction

- CO_2 Reforming a potential reaction of utilising greenhouse gas CO_2
  ⇒ Valuable synthesis gas
- CH_4 well studied feedstock [1]
- Other potential feedstocks sulphur-free GTL (Fischer-Tropsch) fractions not suitable for gasoline
- Catalyst deactivation by carbon deposition major problem [2]
- Highly endothermic

  n-Heptane: ΔH^0_{298K} = +1395 kJ/mol
- Side reactions: the Boudouard and water-gas shift (WGS) reactions

$$C_7H_{16} + 7\ CO_2 \rightleftharpoons 8\ H_2 + 14\ CO$$

Experimental

- Fixed-bed flow reactor
- *n*-Heptane as a model compound (3 mol-%; CO_2: 3 x stoichiometry)
- Thermal experiments (550-800 °C)
- Catalytic experiments (700 and 900 °C, 1-4 bar)
- Commercial 15 wt-% Ni/Al_2O_3 catalyst
- Analysis by two on-line GC's
- Thermodynamics calculated with HSC Chemistry 3.02

Results

Thermodynamic calculations

- Suitable temperature range for experiments from 703 °C to 830 °C
- Pressure effect: p ↑ ⇒ T ↑
- Net stoichiometric equation:
  $C_7H_{16} + 5.98\ CO_2 \rightleftharpoons 11.72\ CO + 5.16\ H_2 + 0.27\ H_2O + 1.29\ CH_4$

  ΔH^0_{298K} = +1084 kJ/mol

Thermal Experiments

- *n*-Heptane cracked to smaller hydrocarbons ethene being the predominant product (Figures 1 and 2)

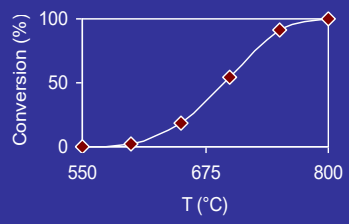


Fig. 1. Conversion of *n*-heptane as a function of temperature.

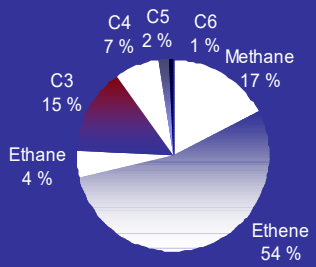


Fig. 2. Product distribution (mol-%) in the thermal experiment at 700 °C.

Catalytic Experiments

- Catalyst deactivated clearly at 700 °C (Figure 3), at 900 °C very slowly
- Higher temperatures are needed at higher pressures
- Obvious occurrence of the side reactions at 900 °C (CO_2 and H_2 consumed + CO formed)
- Experiment at 900 °C product composition compared with thermodynamics: very good agreement
- No thermal cracking at 900 °C

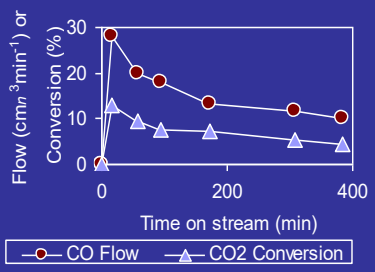


Fig. 3. CO_2 conversion and the outgoing CO flow at 4 bar and 700 °C.

Conclusions

- A promising reaction for the utilisation of CO_2
- High temperatures are required
- Combining the CO_2 reforming with partial oxidation might decrease the need for outside heating and catalyst deactivation

Acknowledgements

The financial support from the Technology Development Centre of Finland (TEKES) and Fortum Oil and Gas Oy is gratefully acknowledged.

References

1. Bradford, M.C.J., Vannice, M.A., CO_2 Reforming of CH_4, *Catal. Rev.-Sci. Eng.* **41**(1) (1999) 1-42.
2. Rostrup-Nielsen, J.R., Bak Hansen, J.-H., CO_2-Reforming of Methane over Transition Metals, *J. Catal.* **144** (1993) 38-49.